Banking Loan Risk Analysis Application

Dataset Details, Schema Dictionary & Preprocessing Guide

1. Dataset Overview & Source

- **Dataset Name:** Default of Credit Card Clients Dataset
- **Source:** UCI Machine Learning Repository (Chung Hua University & Tamkang University, Taiwan)
- **File Format:** Microsoft Excel (.xls / .xlsx) and CSV (.csv)
- **Portfolio Size:** 30,000 customers with 6 months of historical transactions
- **Target Variable:** default payment next month (Binary: 1 = default, 0 = non-default)

2. Attribute Dictionary & Encodings

- **ID:** Customer identifier (1 to 30,000).
- **LIMIT_BAL:** Granted credit limit in NTD (Range: NT0,000 to NT,000,000; Mean: NT67,484).
- **SEX:** Gender (1 = Male [11,888], 2 = Female [18,112]).
- **EDUCATION:** 1 = Graduate School, 2 = University, 3 = High School, 4 = Others, 0/5/6 = Unknown.
- **MARRIAGE:** 1 = Married [13,659], 2 = Single [15,964], 3 = Others [323], 0 = Unknown [54].
- **AGE:** Customer age in years (Range: 21 to 79 years; Mean: 35.5 years).
- **PAY_0 to PAY_6:** Historical repayment statuses for 6 consecutive months (September to April 2005). Scale: -2 = no balance, -1 = paid duly, 0 = revolving credit, 1 to 8 = months of delay.
- **BILL_AMT1 to BILL_AMT6:** Monthly billing statement balances.
- **PAY_AMT1 to PAY_AMT6:** Actual cash payments made across the 6 billing cycles.

3. Preprocessing, Feature Engineering & Batch Workflow

- **Directory Pipeline:** Raw batch files are dropped into data/raw/ and automatically archived into data/processed/ after ingestion.
- **Demographic Categorization:** Derived age_group brackets: 20-29, 30-39, 40-49, 50-59, 60+.
- **Ratios:** Derived utilization_ratio (average bill / limit) and repayment_ratio (total payments / total bills).
- **Delinquency Indicators:** Calculated max_delay_months and delinquent billing cycle counts.